

Forecasting Taxi Demand Across NYC Zones

Background

New York City's taxi industry is a vital part of the city's transportation network, known for its iconic yellow taxis and the more recently introduced green taxis. Yellow taxis are the traditional fleet seen across Manhattan and other boroughs, licensed for street hails and medallion services. In contrast, green taxis (often referred to as Boro Taxis) serve areas outside the core of Manhattan, primarily operating in the outer boroughs such as Queens, Brooklyn, and the Bronx, and have different fare structures and service regulations.

Objective

Develop a forecasting model that predicts the number of yellow taxi trips (or total revenue) for each NYC taxi zone on an hourly basis. This forecast will help optimize yellow taxi allocation, reduce passenger wait times, and improve overall urban mobility in New York City.

Additional Guidelines

1. Candidates are encouraged to supplement the provided dataset with external data sources (e.g., weather information, event calendars) if they believe it can enhance the forecasting model. Please document any additional data used, including where it was obtained and how it was integrated.
2. This problem is open-ended, so you may make any reasonable assumptions as long as they are clearly stated. If certain aspects of the dataset or real-world conditions are unclear, describe the potential limitations or constraints. Demonstrating thoughtful consideration of these factors will be an important part of the evaluation.
3. While there is no single "correct" solution, your work will be assessed based on:
 - a. Clarity of Thought Process: How well you explain your approach, from data exploration to modeling decisions.
 - b. Creativity & Strategy: The extent to which you explore novel ideas, enrich the dataset, and design innovative solutions.
 - c. Rigor in Data Handling: Thoroughness in cleaning, feature engineering, and justifying the methods used.
 - d. Technical Soundness: Suitability of modeling techniques, correctness of implementation, and quality of evaluation metrics.

- e. Communication & Documentation: How effectively you convey your findings, assumptions, and final recommendations.

Deliverables

Please submit the following files as part of your final submission (do not include the original dataset file):

Code Files:

- Python scripts and/or a Jupyter Notebook containing your analysis, feature engineering, modeling, and evaluation.
- If using a Jupyter Notebook, also export it as an HTML file for easier review.

Report:

- A PDF report summarizing your approach, findings, assumptions, limitations, and final recommendations.
- The report should clearly document your thought process, data exploration, feature engineering, model choices, and evaluation metrics.

Data Dictionary

2023 NYC Yellow Taxi Trip Data

This dataset contains detailed trip-level records for NYC yellow taxis in 2023. Each record represents an individual taxi ride and includes key information such as pickup and dropoff timestamps, passenger count, trip distance, fare breakdown, and taxi zone identifiers. The following table is a data dictionary describing each column in the dataset.

Feature	Description	Possible Values
VendorID	A code indicating the TPEP provider that provided the record.	1 = Creative Mobile Technologies, LLC 2 = VeriFone Inc.
tpep_pickup_datetime	The date and time when the meter was engaged.	-
tpep_dropoff_datetime	The date and time when the meter was disengaged.	-
Passenger_count	The number of passengers in the vehicle (driver-entered).	-
Trip_distance	The elapsed trip distance in miles reported by the taximeter.	-
PULocationID	TLC Taxi Zone in which the taximeter was engaged.	-
DOLocationID	TLC Taxi Zone in which the taximeter was disengaged.	-
RateCodeID	The final rate code in effect at the end of the trip.	1 = Standard rate 2 = JFK 3 = Newark 4 = Nassau or Westchester 5 = Negotiated fare 6 = Group ride
Store_and_fwd_flag	Indicates whether the trip record was held in vehicle memory before sending to the vendor ("store and forward").	Y = store and forward trip N = not a store and forward trip
Payment_type	A numeric code signifying how the passenger paid for the trip.	1 = Credit card 2 = Cash 3 = No charge 4 = Dispute 5 = Unknown 6 = Voided trip
Fare_amount	The time-and-distance fare calculated by the meter.	-
Extra	Miscellaneous extras and surcharges (e.g., rush hour and overnight charges).	-
MTA_tax	\$0.50 MTA tax automatically triggered based on the metered rate.	-
Improvement_surcharge	\$0.30 surcharge assessed at the flag drop (introduced in 2015).	-
Tip_amount	Automatically populated for credit card tips (cash tips are not included).	-
Tolls_amount	Total amount of all tolls paid during the trip.	-
Total_amount	The total amount charged to passengers (does not include cash tips).	-
Congestion_Surcharge	Total amount collected in trip for the NYS congestion surcharge.	-
Airport_fee	\$1.25 fee for pickup only at LaGuardia and John F. Kennedy Airports.	-